Assignment Code: DA-AG-007

Statistics Advanced - 2| **Assignment**

Instructions: Carefully read each question. Use Google Docs, Microsoft Word, or a similar tool to create a document where you type out each question along with its answer. Save the document as a PDF, and then upload it to the LMS. Please do not zip or archive the files before uploading them. Each question carries 20 marks.

Total Marks: 180

Question 1: What is hypothesis testing in statistics?

Answer:

Hypothesis testing is a statistical method used to make decisions or draw conclusions about a population based on sample data. It helps researchers determine whether an assumption or claim about a population is supported by evidence.

In hypothesis testing, we first make an assumption called a hypothesis. Then we collect sample data and use statistical calculations to test whether the assumption is valid.

The main steps in hypothesis testing are:

1. State the null hypothesis (H_0) and alternative hypothesis (H_1)
2. Select the significance level (α)
3. Collect sample data
4. Calculate the test statistic
5. Find the p-value
6. Make a decision to reject or fail to reject the null hypothesis

Hypothesis testing is widely used in business, science, medicine, engineering, and research.

Question 2: What is the null hypothesis, and how does it differ from the alternative hypothesis?

Answer:

The **null hypothesis (H_0)** is the initial assumption that there is no significant difference, no effect, or no relationship between variables.

The **alternative hypothesis (H_1 or Ha)** states that there is a significant difference, effect, or relationship.

Example:

Suppose a company claims the average battery life is 10 hours.

Null Hypothesis (H_0): $\mu = 10$
(Battery life is 10 hours)

Alternative Hypothesis (H_1): $\mu \neq 10$
(Battery life is not 10 hours)

Question 3: Explain the significance level in hypothesis testing and its role in deciding the outcome of a test.

Answer:

The significance level, represented by **α (alpha)**, is the probability of rejecting the null hypothesis when it is actually true.

Common values are:

- $\alpha = 0.05$ (5%)
- $\alpha = 0.01$ (1%)
- $\alpha = 0.10$ (10%)

A significance level of 0.05 means there is a 5% chance of making a wrong decision.

Decision Rule:

- If **p-value < α**, reject H_0
- If **p-value > α**, fail to reject H_0

Significance level helps researchers decide whether results are statistically significant.

Question 4: What are Type I and Type II errors? Give examples of each.

Answer:

Type I Error:

This happens when we reject a true null hypothesis.

It is also called a **False Positive**.

Example:

A medical test says a healthy person has a disease.

Type II Error:

This happens when we fail to reject a false null hypothesis.

It is also called a **False Negative**.

Example:

A medical test fails to detect disease in a sick person.

Both errors are important in research and decision-making.

Question 5: What is the difference between a Z-test and a T-test? Explain when to use each.

Answer:

Z-Test:

A Z-test is used when:

- Population standard deviation is known

- Sample size is large ($n > 30$)

It follows the normal distribution.

T-Test:

A T-test is used when:

- Population standard deviation is unknown
- Sample size is small ($n < 30$)

It follows the t-distribution.

Question 6: Write a Python program to generate a binomial distribution with n=10 and p=0.5, then plot its histogram.

(*Include your Python code and output in the code box below.*)

Hint: Generate random number using random function.

Answer:

```
import numpy as np
import matplotlib.pyplot as plt

# Generate binomial data
data = np.random.binomial(n=10, p=0.5, size=1000)

# Plot histogram
plt.hist(data)
plt.title("Binomial Distribution")
plt.xlabel("Values")
plt.ylabel("Frequency")
plt.show()
```

Output Explanation:

This histogram shows the binomial distribution for 1000 random samples.
Most values occur around 5 because probability of success is 0.5.

Question 7: Implement hypothesis testing using Z-statistics for a sample dataset in Python. Show the Python code and interpret the results.

```
sample_data = [49.1, 50.2, 51.0, 48.7, 50.5, 49.8, 50.3, 50.7, 50.2, 49.6,
               50.1, 49.9, 50.8, 50.4, 48.9, 50.6, 50.0, 49.7, 50.2, 49.5,
               50.1, 50.3, 50.4, 50.5, 50.0, 50.7, 49.3, 49.8, 50.2, 50.9,
               50.3, 50.4, 50.0, 49.7, 50.5, 49.9]
```

(*Include your Python code and output in the code box below.*)

Answer:

```
import numpy as np
from scipy.stats import norm
```

```
sample_data = [49.1, 50.2, 51.0, 48.7, 50.5, 49.8, 50.3, 50.7]

sample_mean = np.mean(sample_data)
sample_std = np.std(sample_data, ddof=1)

n = len(sample_data)
mu = 50

z = (sample_mean - mu)/(sample_std/np.sqrt(n))

p_value = 2 * (1 - norm.cdf(abs(z)))

print("Z-statistic =", z)
print("P-value =", p_value)
```

Interpretation:

- If p-value < 0.05 → Reject H_0
- If p-value > 0.05 → Fail to reject H_0

This tells whether the sample mean differs significantly from population mean.

Question 8: Write a Python script to simulate data from a normal distribution and calculate the 95% confidence interval for its mean. Plot the data using Matplotlib.

(*Include your Python code and output in the code box below.*)

Answer:

```
import numpy as np
import matplotlib.pyplot as plt
from scipy import stats

# Generate data
data = np.random.normal(50, 5, 100)

# Confidence interval
mean = np.mean(data)

ci = stats.t.interval(
    0.95,
    len(data)-1,
    loc=mean,
    scale=stats.sem(data)
)

print("95% Confidence Interval:", ci)

# Plot
plt.hist(data)
plt.show()
```

Explanation:

This program:

1. Generates random normal data
2. Calculates mean
3. Finds 95% confidence interval
4. Displays histogram

Confidence interval gives the likely range of the true population mean.

Question 9: Write a Python function to calculate the Z-scores from a dataset and visualize the standardized data using a histogram. Explain what the Z-scores represent in terms of standard deviations from the mean.

(*Include your Python code and output in the code box below.*)

Answer:

```
from scipy.stats import zscore
import matplotlib.pyplot as plt

data = [10, 12, 14, 16, 18]

z_scores = zscore(data)

print("Z-scores:", z_scores)

plt.hist(z_scores)
plt.show()
```

Explanation:

Z-score tells how far a value is from the mean in terms of standard deviations.